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Short proofs of three nonlinear-mean inequalities of Nazarov, Vasyunin and Volberg

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Abstract

For the nonlinear mean $M[a,b] = (a+b)/\sqrt{4+(a-b)^2}$, Nazarov, Vasyunin and Volberg [1] proved several inequalities by computer-assisted positivity tests of large polynomials. We give short computer-free proofs of three of them, I3, I4 and I6 of [1, §§4.3, 4.4, 4.6]. For I3, coordinates given by the hyperbolic distances between $x+i$, $y+i$, $z+i$ in the upper half-plane reduce the inequality to one polynomial identity and the triangle inequality; the same two facts give the reverse inequality when y and z lie on opposite sides of x. For I4 and I6 we use a chart, centred at a value u of M, in which the level set $\{M = u\}$ becomes an antidiagonal line: I4 follows from the concavity of one explicit function, and I6 from the endpoint values of a concave cubic.

1 Results

Throughout, $M[a,b] = (a+b)/\sqrt{4+(a-b)^2}$ for real a, b.

Theorem 1 (I3, [1, §4.3])**.** *Let $x, y, z \geq 0$ with $x(y+z) \leq 2$.*

(a) *If $(x-y)(x-z) \geq 0$, then $M[M[y,x], M[z,x]] \geq M[M[y,z],x]$.*

(b) *If $(x-y)(x-z) \leq 0$, then $M[M[y,x], M[z,x]] \leq M[M[y,z],x]$.*

In particular (a) *holds for $x, y, z \in [0,1]$ with y, z on the same side of x, as stated in [1].*

Theorem 2 (I4, [1, §4.4])**.** *If $0 \leq x \leq y \leq z \leq w \leq 1$, then*

$$M\big[M[x,y], M[z,w]\big] \leq \max\big\{M[x,w], M[y,z]\big\}.$$

Theorem 3 (I6, [1, §4.6])**.** *If $0 \leq z \leq y \leq 1$ and $y - z \leq \frac{3}{4}$, then*

$$M[z,y] \leq M\Big[\frac{z + M[z,y]}{2},\ M\big[M[z,y], y\big]\Big].$$

2 Two lemmas

Lemma 1. *$M[a,b] = M[b,a]$, $M[a,a] = a$, and $\partial_a M[a,b] = 2(2+b^2-ab)/(4+(a-b)^2)^{3/2}$, as $4+(a-b)^2-(a+b)(a-b) = 2(2+b^2-ab)$. Hence $M[\cdot,b]$ is strictly increasing on every interval on which $ab < 2+b^2$. In particular M is strictly increasing in each argument on $[0,1]^2$, so M maps $[0,1]^2$ into $[0,1]$ and $a < M[a,b] < b$ for $0 \leq a < b \leq 1$.*

The chart. Fix $u \in [0,1)$, and let $k = \sqrt{1-u^2}$ and $\varphi = \operatorname{artanh} u$, so that $k\cosh\varphi = 1$ and $u = k\sinh\varphi$. Let $\rho(t) = \operatorname{arsinh}(t/k)$ for $t \in \mathbb{R}$; thus ρ is strictly increasing, $t = k\sinh\rho(t)$ and $\rho(u) = \varphi$. For $d \in \mathbb{R}$ put $\tau = k\tanh d$ and, for $r \in \mathbb{R}$,

$$W_d(r) = 1 + \tau^2\big(\sinh^2 r - \sinh^2\varphi\big), \qquad \psi_d(r) = \operatorname{arsinh}\frac{\sinh r}{\sqrt{W_d(r)}};$$

here $W_d(r) \geq W_d(0) = 1 - u^2\tanh^2 d > 0$.

Lemma 2. (a) *If $\rho(a) = r + d$ and $\rho(b) = r - d$, then $\rho(M[a,b]) = \psi_d(r)$.*

(b) *$\psi_d(\varphi) = \varphi$, and ψ_d is strictly increasing on $\mathbb{R}$ and concave on $[0,\infty)$.*

(c) *For all real a, b, $M[a,b] - u$ has the sign of $\rho(a) + \rho(b) - 2\varphi$.*

Proof. (a) Here $a + b = 2k \sinh r \cosh d$ and $a - b = 2k \cosh r \sinh d$. Since $k^2 \cosh^2 r - 1 = k^2 \sinh^2 r - u^2 = k^2(\sinh^2 r - \sinh^2 \varphi)$,

$$4 + (a-b)^2 = 4 \cosh^2 d \left(1 + \tanh^2 d \,(k^2 \cosh^2 r - 1)\right) = 4 \cosh^2 d \, W_d(r),$$

so $M[a,b] = k \sinh r / \sqrt{W_d(r)}$, which is (a).

(b) $W_d(\varphi) = 1$ gives $\psi_d(\varphi) = \varphi$. Write $W = W_d(r)$, $w_0 = W_d(0)$ and $\psi = \psi_d(r)$, so that $W = w_0 + \tau^2 \sinh^2 r$. Differentiating $\sinh \psi = \sinh r / \sqrt{W}$ gives $\cosh \psi \cdot \psi' = w_0 \cosh r / W^{3/2}$. Moreover $\cosh^2 \psi = (W + \sinh^2 r)/W$ and $W + \sinh^2 r = (1+\tau^2)\cosh^2 r - \tanh^2 d$, because $1 + \tau^2 - w_0 = \tau^2 \cosh^2 \varphi = \tanh^2 d$. Therefore

$$\psi_d'(r) = \frac{w_0}{W_d(r)\sqrt{1 + \tau^2 - \tanh^2 d / \cosh^2 r}} > 0.$$

On $[0,\infty)$ both $W_d(r)$ and $\cosh r$ are nondecreasing, so ψ_d' is nonincreasing there.

(c) Write $\rho(a), \rho(b) = r \pm d$. Since ρ and ψ_d are strictly increasing and $\rho(u) = \varphi = \psi_d(\varphi)$, (a) shows that $M[a,b] - u$, $\psi_d(r) - \psi_d(\varphi)$ and $r - \varphi$ have the same sign. □

In the notation of [1], where $X(u,t) = (u+t)/\sqrt{1-t^2}$, we have $\rho\big(X(u, \tanh\theta)\big) = \varphi + \theta$.

3 Proof of Theorem 1

Fix $x \geq 0$. Since sinh maps $\mathbb{R}$ onto $\mathbb{R}$, we can write

$$y = x + 2\sinh(\sigma + \delta), \qquad z = x + 2\sinh(\sigma - \delta) \qquad (\sigma, \delta \in \mathbb{R}).$$

Put $s = \sinh\sigma$, $h = \sinh\delta$, $g = \cosh\delta$, $c = \cosh 2\delta = g^2 + h^2$, $K = (y+z)/2 = x + 2sg$, and

$$\varepsilon = s^2 - h^2 = \sinh(\sigma+\delta)\sinh(\sigma-\delta) = \tfrac{1}{4}(y-x)(z-x),$$
$$D = s^2 + g^2 = \cosh(\sigma+\delta)\cosh(\sigma-\delta) = c + \varepsilon.$$

Thus case (a) is $\varepsilon \geq 0$ and case (b) is $\varepsilon \leq 0$. Let $L = M[M[y,x], M[z,x]]$, $R = M[M[y,z], x]$ and $v = K/c$.

(i) Since $y - z = 4h\cosh\sigma$ and $1 + 4h^2\cosh^2\sigma = c^2 + 4h^2\varepsilon$ (as $c = 1 + 2h^2$), we have $M[y,z] = K/\sqrt{c^2 + 4h^2\varepsilon}$.

(ii) Since $y - x = 2\sinh(\sigma+\delta)$, we have $M[y,x] = (x + \sinh(\sigma+\delta))/\cosh(\sigma+\delta)$, and similarly for $M[z,x]$. Over the common denominator D, the numerators of $M[y,x] \pm M[z,x]$ are

$$x\big(\cosh(\sigma+\delta) + \cosh(\sigma-\delta)\big) + \sinh 2\sigma = 2\cosh\sigma\,(xg + s),$$
$$x\big(\cosh(\sigma-\delta) - \cosh(\sigma+\delta)\big) + \sinh 2\delta = 2h(g - xs).$$

Hence

$$L^2 = \frac{(1+s^2)(xg+s)^2}{E}, \qquad E = D^2 + h^2(g - xs)^2.$$

(iii) Using $1 + h^2 = g^2$, we get $v + x = 2g(xg+s)/c$ and $v - x = 2(sg - xh^2)/c$, so

$$M[v,x] = \frac{g(xg+s)}{\sqrt{Q}}, \qquad Q = c^2 + (sg - xh^2)^2.$$

(iv) The key identity is

$$(1+s^2)\,Q - g^2 E = \varepsilon\, h^2\,(c - xK). \tag{1}$$

Since $1+s^2-g^2=\varepsilon$, the left side is $\varepsilon Q + g^2(Q-E)$. In $(sg-xh^2)^2 - h^2(g-xs)^2$ the cross terms cancel, leaving $\varepsilon(g^2-x^2h^2)$; and $D^2-c^2=\varepsilon(2c+\varepsilon)$. Hence $Q-E=\varepsilon(g^2-x^2h^2-2c-\varepsilon)$, and the left side of (1) is ε times

$$c^2 - 2cg^2 + g^4 + (sg-xh^2)^2 - \varepsilon g^2 - x^2g^2h^2 = h^4 + g^2h^2 - 2xsgh^2 - x^2h^2 = h^2(c-xK),$$

by $c-g^2=h^2$, $s^2-\varepsilon=h^2$, $g^2-h^2=1$ and $c=g^2+h^2$.

Step 1. By (ii)–(iv),

$$L^2 - M[v,x]^2 = \frac{(xg+s)^2\,\varepsilon\,h^2\,(c-xK)}{EQ},$$

where $E,Q>0$ and $c\ge 1\ge \frac12 x(y+z)=xK$. Since $x,y,z,v\ge 0$, both L and $M[v,x]$ are ≥ 0. So $L\ge M[v,x]$ in case (a) and $L\le M[v,x]$ in case (b).

Step 2. By (i), $M[y,z]\le v$ in case (a) and $M[y,z]\ge v$ in case (b).

Step 3. Both $M[y,z]$ and v lie in $[0,K]$, since $c^2+4h^2\varepsilon = 1+4h^2\cosh^2\sigma \ge 1$ and $c\ge 1$. For $0\le a\le K$ we have $ax\le Kx\le 1<2+x^2$, so $M[\cdot,x]$ is increasing on $[0,K]$ by Lemma 1. Hence $L\ge M[v,x]\ge M[M[y,z],x]=R$ in case (a), and $L\le M[v,x]\le R$ in case (b). □

Remark. For $P_t = t+i$ in the upper half-plane, $\sinh\frac12 d(P_a,P_b) = \frac12|a-b|$; so (i) reads $\cosh\frac12 d(P_y,P_z)=\sqrt{c^2+4h^2\varepsilon}$, while $c=\cosh\frac12|d(P_x,P_y)\mp d(P_x,P_z)|$ in case (a), resp. (b), and Step 2 is the hyperbolic triangle inequality.

4 Proof of Theorem 2

By Lemma 1, all values of M below lie in $[0,1]$, where M is increasing in each argument. Let $u=\max\{M[x,w],M[y,z]\}$. If $u=1$ there is nothing to prove, so let $u<1$.

Reduction. If $M[x,w]<u$, then $M[x,w]<u=M[y,z]\le M[y,w]$, so $M[x',w]=u$ for some $x'\in(x,y]$. If $M[y,z]<u$, then $M[y,z]<u=M[x,w]\le M[y,w]$, so $M[y,z']=u$ for some $z'\in(z,w]$. Replacing x by x', or z by z', preserves $x\le y\le z\le w$ and u, and does not decrease $M[M[x,y],M[z,w]]$. So we may assume that $M[x,w]=M[y,z]=u$; then $y\le M[y,z]=u$.

The critical case. Take the chart at u. By Lemma 2(c), $\rho(x)+\rho(w)=2\varphi=\rho(y)+\rho(z)$. Write $\rho(x),\rho(y)=\varphi-\lambda\mp d$; then $\rho(w),\rho(z)=\varphi+\lambda\pm d$, and $0\le x\le y\le u$ gives $0\le d\le\lambda\le\varphi$. So the pairs (x,y) and (z,w) have chart midpoints $\varphi\mp\lambda\in[0,\infty)$ and the same half-difference d, and by Lemma 2(a),(b)

$$\rho\big(M[x,y]\big)+\rho\big(M[z,w]\big)=\psi_d(\varphi-\lambda)+\psi_d(\varphi+\lambda)\le 2\psi_d(\varphi)=2\varphi.$$

By Lemma 2(c), $M[M[x,y],M[z,w]]\le u$. □

The reduction in [1, §4.4] raises x and y, starting from $x\le y\le u\le z\le w$; but $u\le z$ can fail (e.g. $x=y=z=0<w$). Raising x or z, as above, avoids this.

5 Proof of Theorem 3

Let $u=M[z,y]$. If $z=y$, both sides equal y. Otherwise $z<u<y\le 1$ by Lemma 1, and we take the chart at u. By Lemma 2(c), $\rho(z)+\rho(y)=2\varphi$; write $\rho(z),\rho(y)=\varphi\mp 2\beta$ with $\beta>0$. Then $y-z=2k\cosh\varphi\sinh 2\beta=2\sinh 2\beta$, so $\sinh 2\beta\le\frac38$, and $z\ge 0$ means $\varphi\ge 2\beta$. Let $A=(z+u)/2$,

$B = M[u,y]$, $S = \sinh\beta$ and $C = \cosh\beta$. Then $A/k = \frac12(\sinh(\varphi-2\beta)+\sinh\varphi) = C\sinh(\varphi-\beta)$. Since $\rho(u), \rho(y) = (\varphi+\beta)\mp\beta$ and $\sinh^2(\varphi+\beta) - \sinh^2\varphi = S\sinh(2\varphi+\beta)$, Lemma 2(a) gives

$$B/k = \frac{\sinh(\varphi+\beta)}{\sqrt{W}}, \qquad W = W_\beta(\varphi+\beta) = 1+\tau^2 S\sinh(2\varphi+\beta) \ge 1, \qquad \tau = k\tanh\beta.$$

By Lemma 2(c), the claim $M[A,B] \ge u$ means $\rho(A)+\rho(B) \ge 2\varphi$, i.e. $\sinh(\rho(A)+\rho(B)) \ge \sinh 2\varphi$. Now $\sinh\rho(A) = A/k \ge \sinh(\varphi-\beta) \ge 0$, $\cosh\rho(A) \ge \cosh(\varphi-\beta)$, $\sinh\rho(B) = B/k \ge 0$ and $\cosh\rho(B) = \sqrt{1+B^2/k^2} \ge \cosh(\varphi+\beta)/\sqrt{W}$. Hence

$$\begin{aligned}\sqrt{W}\sinh\bigl(\rho(A)+\rho(B)\bigr) &\ge C\sinh(\varphi-\beta)\cosh(\varphi+\beta) + \cosh(\varphi-\beta)\sinh(\varphi+\beta)\\ &= \sinh 2\varphi + (C-1)\sinh(\varphi-\beta)\cosh(\varphi+\beta).\end{aligned}$$

On the other hand, $\sqrt{W} \le (1+W)/2$ and $k^2\sinh 2\varphi = 2\tanh\varphi$ give $\sqrt{W}\sinh 2\varphi \le \sinh 2\varphi + \tanh\varphi\tanh^2\beta\, S\sinh(2\varphi+\beta)$. So it suffices that

$$(C-1)\sinh(\varphi-\beta)\cosh(\varphi+\beta) \ge \tanh\varphi\tanh^2\beta\, S\sinh(2\varphi+\beta).$$

Since $C-1 = C^2\tanh^2\beta/(1+C)$ and $2\sinh(\varphi-\beta)\cosh(\varphi+\beta) = \sinh 2\varphi - \sinh 2\beta$, this is $F(\varphi) \ge 0$, where

$$F(\varphi) = C^2(\sinh 2\varphi - \sinh 2\beta) - 2S(1+C)\tanh\varphi\,\sinh(2\varphi+\beta).$$

Let $t = \tanh\varphi \in [\tanh 2\beta, 1)$. As $(1-t^2)\sinh 2\varphi = 2t$ and $(1-t^2)\cosh 2\varphi = 1+t^2$, we have $(1-t^2)F(\varphi) = P(t)$, where

$$P(t) = C^2\bigl(2t - (1-t^2)\sinh 2\beta\bigr) - 2tS(1+C)\bigl(2tC + (1+t^2)S\bigr).$$

We shall use $S \le SC = \frac12\sinh 2\beta \le \frac{3}{16}$, $\tanh\beta \le S$, $C^2 = 1+S^2 < 1.21$ and $5\beta \le \frac52\sinh 2\beta \le \frac{15}{16}$. The coefficients of t^3 and t^2 in P are $-2S^2(1+C)$ and $2SC(C^2-2C-2)$, both negative. So P is concave on $[0,\infty)$, and it suffices to show that $P(1) > 0$ and $P(\tanh 2\beta) > 0$. First,

$$P(1) = 2C^2 - 2\sinh 2\beta\,(1+C)(1+\tanh\beta) > 2 - \tfrac34\cdot 2.1\cdot 1.2 > 0.$$

Second, $P(\tanh 2\beta) = (1-\tanh^2 2\beta)F(2\beta)$, and $\sinh 4\beta - \sinh 2\beta = 2S\cosh 3\beta$, so

$$F(2\beta) = 2S\bigl(C^2\cosh 3\beta - (1+C)\tanh 2\beta\sinh 5\beta\bigr) \ge 2SC^2\bigl(1 - 2\tanh 2\beta\sinh 5\beta\bigr) > 0,$$

because $1+C \le 2C^2$, $\cosh 3\beta \ge 1$, and $2\tanh 2\beta\sinh 5\beta < \frac34\sinh 1 < 1$ (as $e < 3$ gives $\sinh 1 < \frac12(3-\frac13) = \frac43$). □

References

[1] F. Nazarov, V. Vasyunin, A. Volberg, *On a Bellman function associated with the Chang–Wilson–Wolff theorem: a case study*, arXiv:2204.01336.